COMPANYVAL AI

AI-Assisted Business Valuation

ICAI AICA Level 2 — Capstone Project Documentation

Project Type: AI-Assisted Financial Decision Support System
Domain: Business Valuation / Financial Analysis
Application Name: CompanyVal AI
Tagline: *Upload. Understand. Question. Simulate. Value.*

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1. EXECUTIVE SUMMARY

Business valuation is traditionally a combination of financial analysis, professional judgement, business understanding, forecasting and application of suitable valuation methodologies. A major challenge is that historical financial statements alone do not provide sufficient information regarding the sustainability of growth, customer concentration, promoter dependency, recurring revenues, capital expenditure requirements, business risks and other qualitative factors that materially influence valuation.

CompanyVal AI has been conceptualised as an AI-assisted business valuation platform that combines:

- financial-statement extraction using Python;
- visual verification using a multimodal AI model;
- deterministic financial and valuation calculations;
- a configurable financial rules engine;
- an adaptive AI-driven business interview;
- human approval of material assumptions;
- scenario and sensitivity analysis;
- explainable valuation outputs; and
- an AI-assisted professional valuation report.

The system accepts up to three years of financial statements and converts them into a structured financial dataset.

Python libraries first extract financial information from the uploaded documents. Relevant pages are then rendered as high-resolution images and independently examined by a multimodal AI model. The Python-extracted data and AI-verified data are reconciled before the user approves the historical financial information.

CompanyVal AI subsequently calculates historical financial ratios, identifies anomalies and activates predefined financial rules. These rules determine what additional information is required from the user.

Instead of presenting a static questionnaire, the system conducts an **adaptive AI Interview**. Questions are generated according to the company's financial characteristics and previous answers.

After obtaining the required information, the deterministic valuation engine calculates values using:

1. Discounted Cash Flow Method;
2. Market Multiple Method; and
3. Adjusted Net Asset Value Method.

Users can subsequently alter major assumptions within the **Simulation Lab** and observe the immediate impact on company value.

Artificial Intelligence is deliberately separated from the authoritative mathematical valuation engine. AI assists with document interpretation, adaptive questioning, qualitative analysis and professional report writing, whereas financial calculations remain deterministic and reproducible.

The final output is therefore an **AI-Assisted Indicative Business Valuation**, rather than an autonomous AI opinion on value.

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2. PROBLEM STATEMENT

Business valuation requires more than applying a formula to accounting numbers.

Historical financial statements may indicate:

- turnover;
- profitability;
- debt;
- assets;
- cash flows; and
- working capital.

However, they normally do not adequately answer questions such as:

- Is recent revenue growth sustainable?
- Was a major contract non-recurring?
- How concentrated is the customer base?
- Is the company dependent on its promoters?
- Are current margins sustainable?
- Will substantial future capital expenditure be required?
- Is increasing working capital affecting cash generation?
- Are there significant contingent liabilities?
- What level of future growth is realistically achievable?
- Are historical profits representative of maintainable earnings?

A conventional valuation calculator cannot intelligently obtain this missing information.

Conversely, asking a generative AI model to directly read financial statements and determine the value of the company introduces significant risks:

- incorrect extraction;
- hallucinated values;
- non-reproducible calculations;
- unsupported assumptions;
- lack of audit trail; and
- lack of explainability.

The project therefore addresses the following question:

> **How can Artificial Intelligence, Python-based financial statement extraction and deterministic valuation rules be combined to understand a business interactively and generate an explainable indicative valuation range?**

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3. PROJECT OBJECTIVE

The primary objective of CompanyVal AI is:

> **To develop an AI-assisted business valuation simulator capable of analysing three years of financial statements, intelligently collecting valuation-relevant qualitative information and applying multiple deterministic valuation methodologies to produce an explainable indicative valuation range.**

Secondary objectives are to demonstrate the practical use of:

- generative AI;
- multimodal document analysis;
- Python automation;
- financial-data extraction;
- financial ratio analysis;
- rule-based systems;
- adaptive AI questioning;
- structured AI responses;
- human-in-the-loop decision making;
- business valuation techniques;
- sensitivity analysis;
- scenario modelling;
- explainable AI;
- secure API integration; and
- automated professional reporting.

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4. SCOPE OF THE PROJECT

4.1 Included Scope

CompanyVal AI is designed to support:

Financial Inputs

- Balance Sheet
- Statement of Profit & Loss
- Cash Flow Statement
- selected Notes to Accounts

File Formats

- PDF
- XLSX
- XLS

Historical Period

Up to three financial years.

Financial Analysis

The application calculates, wherever sufficient information is available:

- revenue growth;
- revenue CAGR;
- EBITDA margin;
- PAT margin;
- ROE;
- ROCE;
- current ratio;
- debt/equity;
- debt/EBITDA;
- asset turnover;
- receivable days;
- inventory days;
- payable days;
- cash conversion cycle;
- CFO/PAT;
- earnings volatility; and
- working-capital trends.

Valuation Methods

1. Discounted Cash Flow
2. Market Multiple
3. Adjusted Net Asset Value

Additional Functionality

- AI Interview;
- normalisation adjustments;
- Bear/Base/Bull scenarios;
- sensitivity analysis;
- valuation explainability;
- AI insights; and
- professional PDF reporting.

4.2 Excluded Scope

The capstone prototype does not attempt to automatically issue:

- a Registered Valuer certificate;
- a statutory valuation report;
- an audit opinion;
- a fairness opinion;
- a merchant banker certificate; or
- any legally mandated professional certification.

CompanyVal AI is a **decision-support and valuation simulation platform**.

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5. PROPOSED SOLUTION

The complete application workflow is:

```
CREATE VALUATION CASE
          ↓
UPLOAD THREE YEARS' FINANCIAL STATEMENTS
          ↓
PYTHON DOCUMENT EXTRACTION
          ↓
RENDER RELEVANT PDF PAGES
          ↓
MULTIMODAL AI VISUAL VERIFICATION
          ↓
```

```
PYTHON ↔ AI RECONCILIATION
          ↓
ACCOUNTING VALIDATION
          ↓
HUMAN REVIEW & APPROVAL
          ↓
LOCK VERIFIED HISTORICAL FINANCIALS
          ↓
FINANCIAL ANALYTICS
          ↓
RULES ENGINE
          ↓
ADAPTIVE AI INTERVIEW
          ↓
NORMALISATION & ASSUMPTION REVIEW
          ↓
VALUATION ENGINE
          ↓
DCF + MARKET MULTIPLE + ADJUSTED NAV
          ↓
SIMULATION & SENSITIVITY ANALYSIS
          ↓
AI INSIGHTS
          ↓
AI-ASSISTED PROFESSIONAL REPORT
```

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6. KEY INNOVATION OF COMPANYVAL AI

The principal innovation is not merely the use of an LLM.

The project creates a controlled architecture in which different technologies perform the tasks for which they are most suitable.

Python performs:

- PDF processing;
- table and text extraction;
- financial-data standardisation;
- accounting checks;
- financial ratios;
- valuation calculations;
- sensitivity analysis;
- scenario recalculation; and
- deterministic rules.

Artificial Intelligence performs:

- visual verification;
- semantic interpretation;
- contextual question generation;
- classification of qualitative answers;
- identification of business risks;
- explanation of financial observations; and
- professional report narration.

The user performs:

- discrepancy resolution;
- approval of historical financials;
- confirmation of assumptions;
- acceptance/rejection of AI recommendations; and
- professional judgement.

This creates:

> **AI + Rules + Financial Mathematics + Human Oversight**

rather than:

> **AI = Valuation**

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7. TECHNOLOGY ARCHITECTURE

7.1 Frontend

The proposed frontend uses:

- React;
- Vite;
- TypeScript;
- Tailwind CSS;
- reusable UI components;
- charting components; and
- responsive PWA architecture.

7.2 Backend

The backend uses:

- Python;
- FastAPI;
- SQLAlchemy;
- Pydantic;
- PostgreSQL; and
- REST APIs.

SQLite may be used during local prototype development.

7.3 Document Intelligence

The document-processing layer uses suitable Python libraries including:

- PyMuPDF;
- PyMuPDF4LLM;
- pandas;
- NumPy;
- Pillow;
- openpyxl; and
- optional fallback extraction mechanisms.

7.4 AI Layer

The application is designed with a configurable AI provider.

The project configuration presently specifies:

Provider: Google Gemini
Default model configured for the project: Gemini 3.6 Flash

The AI model is accessed only from the backend.

---

8. SYSTEM ARCHITECTURE

```
                    USER
                      │
```

```
                   ▼
             COMPANYVAL AI PWA
                   │
                   ▼
               FASTAPI BACKEND
                   │
      ┌────────────┼────────────┐
      │            │            │
      ▼            ▼            ▼
DOCUMENT ENGINE   FINANCIAL ENGINE   AI ENGINE
      │            │            │
      │            │            │
  PyMuPDF       Python Rules     Gemini API
PyMuPDF4LLM     Ratios
               Valuation
      │            │            │
      └────────────┼────────────┘
                   ▼
                DATABASE
                   │
                   ▼
             REPORT ENGINE
```

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9. FINANCIAL DOCUMENT INTELLIGENCE

One of the central components of CompanyVal AI is its document-intelligence pipeline.

The application does not rely solely on an LLM to extract financial data.

Instead, two independent methods are used.

9.1 Python Extraction

When a financial statement is uploaded, the backend:

1. validates the file;
2. generates a document identifier;
3. calculates a file hash;
4. examines PDF text and page structure;
5. detects probable financial statements;
6. extracts tables and financial line items;
7. associates values with financial years; and

8. stores the source page for each material number.

9.2 PDF Page Rendering

Relevant financial-statement pages are rendered into high-resolution images.

Typical examples include:

- Balance Sheet page;
- Profit & Loss page;
- Cash Flow Statement page; and
- material supporting Notes.

These page images are supplied to the multimodal AI verification layer.

9.3 Dual Verification

The architecture can be represented as:

```
                   PDF
                /
               /
     PYTHON PARSER    PAGE IMAGE
           │               │
           │               ▼
           │         MULTIMODAL AI
           │               │
           └───────┬───────┘
                   ▼
           RECONCILIATION
```

For every material financial amount, the system may store:

- Python-extracted value;
- AI-verified value;
- confidence;
- source page; and
- verification status.

9.4 Verification Status

Possible results include:

Verified

Python and AI readings agree.

Difference

The two readings differ.

Ambiguous

The source document does not permit confident interpretation.

Missing

The required value could not be established.

9.5 Human Review

Where a discrepancy exists, the user is presented with:

```
Python Extraction       ₹36.20 Lakh
AI Visual Verification  ₹38.20 Lakh

Status                   NEEDS REVIEW
Source                    Page 18
```

The user selects or manually enters the correct amount.

The selected value becomes the authorised financial amount.

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10. DATA TRACEABILITY

Each important financial number can be stored together with its source information.

Conceptually:

```json
{
  "metric": "revenue",
  "period": "FY2025-26",
  "approved_value": 124500000,
  "source_page": 18,
  "python_value": 124500000,
  "ai_verified_value": 124500000,
  "verification_status": "verified"
}
```

This allows the application to explain:

> **Where did this number come from?**

This is particularly important when AI is used in financial applications.

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11. FINANCIAL DATA STANDARDISATION

Different companies may use different descriptions for similar financial items.

For example:

```
Sales
Turnover
Operating Revenue
Revenue from Operations
```

are internally mapped to:

```
revenue
```

Likewise:

```
Net Profit
Profit After Tax
Profit for the Year
```

are mapped to:

```
PAT
```

This canonical financial schema allows subsequent calculations to work independently of the company's original terminology.

---

12. ACCOUNTING VALIDATION

Before financial information is used for valuation, deterministic accounting checks are performed.

Examples include:

Balance Sheet Check

```
Total Assets ≈ Equity + Liabilities
```

Profit Check

```
Profit Before Tax - Tax ≈ Profit After Tax
```

Cash Flow Check

```
Opening Cash
+ Net Cash Movement
≈ Closing Cash
```

Comparative-Figure Check

Where the current financial statement contains comparative figures for the previous year, those values may be compared against the previously uploaded financial statement.

Material differences are flagged for review.

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13. FINANCIAL ANALYTICS

After financial information has been approved, CompanyVal AI calculates analytical metrics.

Examples include:

Growth

```
Revenue Growth =
(Current Revenue - Previous Revenue)
÷ Previous Revenue
```

Revenue CAGR

```
CAGR =
(Ending Revenue ÷ Beginning Revenue)^(1/n) - 1
```

EBITDA Margin

```
EBITDA Margin =
EBITDA ÷ Revenue
```

Debt-Equity Ratio

```
Debt / Equity =
Total Interest-Bearing Debt ÷ Net Worth
```

Cash Conversion Indicator

```
CFO / PAT =
Operating Cash Flow ÷ Profit After Tax
```

These ratios become inputs to both:

- the rules engine; and
- the AI Interview.

---

14. RULES ENGINE

The rules engine acts as the deterministic intelligence layer between financial analysis and generative AI.

Example:

```
IF Revenue Growth > 25%
THEN investigate sustainability of growth
```

The rule does not automatically change the valuation.

Instead, it determines what information should be obtained.

14.1 Illustrative Rules

Strong Growth

```
IF YoY Revenue Growth > 25%
THEN investigate growth sustainability
```

Revenue Decline

```
IF Revenue Decline > 10%
THEN investigate cause and recovery
```

Margin Movement

```
IF EBITDA Margin changes > 3 percentage points
THEN investigate margin movement
```

Cash Conversion

```
IF CFO / PAT < 0.70
THEN investigate working capital and earnings quality
```

Leverage

```
IF Debt / Equity > 1.50
THEN investigate leverage and refinancing
```

Customer Concentration

```
IF Largest Customer > 25% of Revenue
THEN flag concentration risk
```

The thresholds are application rules and do not represent statutory valuation standards.

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15. AI INTERVIEW

A major differentiator of CompanyVal AI is its adaptive interview.

It does not display the same questionnaire for every business.

The question planner receives:

```
Verified Financial Data
+
Financial Ratios
+
Triggered Rules
+
Previous Answers
+
Missing Valuation Inputs
```

and determines the next relevant question.

15.1 Example

Assume:

```
FY 2024-25 Revenue = ₹9.02 Cr
FY 2025-26 Revenue = ₹13.00 Cr
Growth             = 44%
```

The growth rule is triggered.

CompanyVal AI asks:

What principally contributed to the significant increase in revenue during FY 2025-26?

Possible responses:

- New Customers
- Price Increase
- New Geography
- New Product
- One-Time Order
- Acquisition
- Other

If the user selects:

One-Time Order

the next question may be:

Approximately what amount of FY 2025-26 revenue arose from this order?

The subsequent question may be:

Do you reasonably expect similar orders to recur?

This creates a contextual interview instead of a static questionnaire.

---

16. QUESTION PRIORITY

Potential questions are prioritised according to factors such as:

```
Materiality
×
Valuation Impact
×
Uncertainty
=
Question Priority
```

This enables the application to concentrate on the information that may materially affect value.

The objective is therefore:

Fewer relevant questions rather than a large generic questionnaire.

17. HUMAN-IN-THE-LOOP AI

AI recommendations do not become valuation assumptions automatically.

For example:

```
Current Growth Assumption       18%
AI Suggested Growth             13%

Reason:
Recent revenue growth included a
material non-recurring contract.
```

The user may select:

```
KEEP 18%
ACCEPT 13%
ENTER ANOTHER VALUE
```

Only the explicitly approved assumption is used.

This prevents generative AI from silently modifying valuation assumptions.

18. NORMALISATION OF EARNINGS

The system permits explicit normalisation adjustments.

Examples include:

- one-time revenue;
- exceptional expenses;
- extraordinary losses;
- unusually high promoter remuneration;
- non-operating income;
- litigation expenditure;
- abnormal related-party transactions.

Each adjustment retains:

- reported amount;
- adjustment amount;

- normalised amount;
- reason; and
- user approval.

---

19. VALUATION METHODOLOGIES

CompanyVal AI uses three principal approaches.

19.1 DISCOUNTED CASH FLOW

A five-year FCFF model may be developed.

The basic flow is:

```
Revenue
    ↓
EBITDA
    ↓
EBIT
    ↓
NOPAT
    ↓
+ Depreciation
- Capital Expenditure
- Increase in Net Working Capital
    ↓
FCFF
```

19.2 NOPAT

```
NOPAT =
EBIT × (1 - Tax Rate)
```

19.3 FREE CASH FLOW TO FIRM

```
FCFF =
NOPAT
+ Depreciation
- Capital Expenditure
- Change in Net Working Capital
```

19.4 PRESENT VALUE

```
PV of FCFF =
FCFFt
────────────
(1 + WACC)^t
```

19.5 TERMINAL VALUE

Using a perpetual growth approach:

```
Terminal Value =
FCFF(n+1)
────────────
WACC - g
```

where:

- `WACC` = Weighted Average Cost of Capital
- `g` = Terminal Growth Rate

The application must prevent economically invalid combinations such as:

```
Terminal Growth ≥ WACC
```

19.6 ENTERPRISE VALUE

```
Enterprise Value =
PV of Forecast FCFF
+
PV of Terminal Value
```

19.7 EQUITY VALUE

```
Equity Value =
Enterprise Value
- Debt
+ Cash & Cash Equivalents
```

19.8 PER-SHARE VALUE

Where the number of diluted shares is available:

```
Value Per Share =
Equity Value ÷ Diluted Shares
```

---

20. MARKET MULTIPLE METHOD

CompanyVal AI can support:

- EV/EBITDA;
- P/E; and
- EV/Revenue.

For example:

```
Enterprise Value =
Normalised EBITDA
```

```
×
Selected EV/EBITDA Multiple
```

AI is not permitted to invent a comparable-company multiple.

The multiple must originate from:

- configured industry benchmark data;
- explicit user input; or
- another authorised source.

---

21. ADJUSTED NET ASSET VALUE

The Asset Approach allows selected balance-sheet values to be adjusted.

Examples include:

- land;
- buildings;
- investments;
- inventory;
- receivables;
- intangible assets;
- contingent liabilities.

The calculation is broadly:

```
Adjusted Asset Value
-
Outside Liabilities
=
Adjusted Net Asset Value
```

Each adjustment is separately disclosed.

---

22. VALUATION TRIANGULATION

Instead of presenting one unexplained AI-generated number, the application compares values produced using different methodologies.

For example:

Method	Illustrative Value
DCF	₹22.40 Cr
Market Multiple	₹20.80 Cr
Adjusted NAV	₹17.90 Cr

A weighted central estimate may then be calculated using disclosed methodology weights.

Illustratively:

```
DCF               50%
Market Multiple   30%
Adjusted NAV      20%
```

Therefore:

```
Central Estimate =
Σ(Method Value × Method Weight)
```

The weighting remains transparent and user-configurable.

---

23. INDICATIVE VALUATION RANGE

CompanyVal AI presents an:

Indicative Valuation Range

instead of claiming one mathematically exact company value.

For the basic capstone implementation:

```
Lower Bound =
Lowest selected methodology value

Upper Bound =
Highest selected methodology value
```

Scenario ranges may be displayed separately.

---

24. VALUATION READINESS

CompanyVal AI displays a Valuation Readiness indicator.

An illustrative weighting is:

Component	Weight
Financial Data Completeness	25%
Financial Verification	20%
Forecast Inputs	20%
AI Interview Completion	20%
Risk Information	15%
Total	**100%**

This indicates whether sufficient information has been supplied to conduct the analysis.

It does **not** claim the resulting valuation is 94% accurate.

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25. VALUATION CONFIDENCE

Valuation Confidence is separately assessed based on indicators such as:

- verification quality;
- forecast completeness;
- methodology consistency;
- normalisation completeness;
- assumption sensitivity; and
- degree of agreement between methods.

Possible qualitative classifications include:

- High Confidence
- Moderate Confidence
- Low Confidence

Again, this is an analytical confidence indicator rather than a statistical assurance statement.

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26. SIMULATION LAB

The Simulation Lab allows the user to change major valuation drivers.

Examples include:

- Revenue Growth;
- EBITDA Margin;
- WACC;
- Terminal Growth Rate;
- EV/EBITDA Multiple;
- Capital Expenditure; and
- Working-Capital assumptions.

As the values change, the valuation is recalculated immediately.

Example:

```
Revenue CAGR       +2%
Valuation Impact   +₹0.28 Cr
```

The simulation calculation is performed by the valuation engine.

An AI call is not required each time a slider moves.

---

27. SCENARIO ANALYSIS

The system supports:

Bear Case

Conservative assumptions.

Base Case

Accepted management/base assumptions.

Bull Case

Optimistic but explicit assumptions.

The results may be displayed simultaneously to illustrate how changes in business conditions affect company value.

28. SENSITIVITY ANALYSIS

CompanyVal AI identifies which assumptions have the greatest impact on valuation.

Examples include:

- WACC vs Terminal Growth matrix;
- Tornado analysis;
- assumption impact table.

Example:

```
Assumption                Change       Valuation Impact

Revenue Growth            +2%          +₹0.28 Cr
EBITDA Margin              +2%          +₹0.22 Cr
WACC                       -1%          +₹0.17 Cr
Exit Multiple              +1.0x        +₹0.23 Cr
Terminal Growth            +0.5%        +₹0.09 Cr
```

This helps explain not only:

> **What is the valuation?**

but also:

> **What is driving the valuation?**

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29. EXPLAINABLE AI

CompanyVal AI incorporates an explainability layer.

A user can view:

Positive Drivers

For example:

- strong historical growth;
- improving operating margins;
- recurring revenue;

- strong cash generation;
- low leverage.

Negative Drivers

For example:

- customer concentration;
- increasing receivable days;
- promoter dependency;
- working-capital intensity;
- regulatory exposure.

Most Sensitive Assumption

The application can identify which assumption currently creates the greatest movement in value.

This creates an explainable decision-support system instead of an opaque AI answer.

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30. AI INSIGHTS

The AI Insights module may analyse structured information and provide observations concerning:

- financial health;
- earnings quality;
- growth quality;
- business strengths;
- key risks;
- assumption reasonableness; and
- major valuation drivers.

AI insights must be grounded exclusively in information available within the valuation case.

The model must not fabricate company-specific facts.

---

31. PROFESSIONAL REPORT GENERATION

After all calculations are complete, the structured case data is supplied to the AI report-generation layer.

The AI receives:

```
Company Profile
+
Verified Historical Financials
+
Financial Ratios
+
Normalisation Adjustments
+
Interview Answers
+
Approved Assumptions
+
Risks
+
DCF Result
+
Market Multiple Result
+
Adjusted NAV Result
+
Sensitivity Analysis
+
Scenario Analysis
```

The AI then prepares the narrative components of the report.

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32. CRITICAL AI REPORT RULE

The report-generation model is instructed:

> **All numerical financial and valuation results supplied to you are authoritative outputs of CompanyVal AI's deterministic financial engine. You may explain, contextualise and compare those results, but you must not alter, fabricate, substitute or silently recalculate them.**

This is a fundamental safeguard of the project.

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33. REPORT STRUCTURE

The final professional report may contain:

1. Cover Page

2. Executive Summary
3. Company Profile
4. Documents Analysed
5. Historical Financial Performance
6. Financial Ratio Analysis
7. Earnings Quality
8. Normalisation Adjustments
9. Business Assessment
10. AI Interview Findings
11. Key Risks
12. Valuation Assumptions
13. DCF Valuation
14. Market Multiple Valuation
15. Adjusted NAV
16. Method Comparison
17. Sensitivity Analysis
18. Scenario Analysis
19. Key Value Drivers
20. Indicative Valuation Range
21. Conclusion
22. Assumptions
23. Disclaimer
24. Appendices and Data Sources

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34. AI GOVERNANCE

CompanyVal AI follows several controls designed to reduce AI-related risk.

34.1 AI Does Not Calculate the Authoritative Valuation

The mathematical engine is deterministic.

34.2 AI Cannot Silently Change Historical Financials

Historical values are locked after user approval.

34.3 AI Recommendations Require Human Acceptance

Suggested assumptions remain recommendations until approved.

34.4 AI Output Uses Structured Schemas

Where possible, AI responses are requested in machine-readable structured form.

34.5 Source Data Is Retained

Important financial numbers retain traceability to the source document.

34.6 AI Failure Does Not Destroy the Workflow

If the AI service is unavailable:

- Python extraction can remain available;
- user review remains available;
- historical data remains stored; and
- deterministic financial calculations remain independent.

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35. API SECURITY

The Gemini API is never called directly from frontend JavaScript.

Correct architecture:

```
Browser
    ↓
CompanyVal AI Backend
    ↓
AI Provider
```

The API credential remains on the server side.

35.1 API KEY HANDLING

When the API key is first configured:

1. user enters the key;
2. backend verifies connectivity;
3. key is encrypted;
4. encrypted value is stored;
5. frontend subsequently receives only masked status information.

Example:

```
Gemini API Key

••••••••••••••••••••

Status: Connected
```

The application does not provide a button to reveal the stored key.

A user can replace the key where necessary.

---

36. DATABASE DESIGN

Core entities include:

```
Users
Companies
Valuation Cases
Documents
Document Pages
Financial Periods
Financial Line Items
Extraction Results
Verification Results
Financial Ratios
Normalisation Adjustments
Rule Triggers
AI Interview Sessions
Questions
Answers
Valuation Assumptions
Valuation Runs
Method Results
Scenario Runs
AI Insights
Reports
AI Call Logs
Application Settings
Audit Logs
```

The database therefore maintains separation between:

- source information;
- extracted information;

• verified information;
• approved assumptions; and
• valuation outputs.

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37. AUDIT TRAIL

The application records significant actions such as:

• document upload;
• extraction completion;
• AI verification;
• manual correction;
• approval of financial data;
• financial lock/unlock;
• assumption changes;
• AI recommendation acceptance;
• valuation run;
• scenario run;
• report generation; and
• AI configuration changes.

Sensitive credentials are excluded from audit logs.

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38. USER INTERFACE

The application comprises the following principal modules:

1. Dashboard
2. New Valuation
3. Financials
4. AI Interview
5. Valuations
6. Simulation Lab
7. AI Insights
8. Reports
9. Settings

The interface uses a professional financial-SaaS visual style with:

• clean white backgrounds;
• soft blue accents;
• subtle teal/green positive indicators;
• orange warnings;

• restrained risk colours;
• lightweight cards;
• analytical charts; and
• clear financial hierarchy.

---

39. DASHBOARD

The Dashboard provides an overview of:

• valuation cases;
• completed valuations;
• cases in progress;
• average valuation;
• valuation readiness;
• recent cases;
• valuation trends;
• method comparison;
• simulation summary;
• key insights; and
• report preview.

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40. FINANCIALS MODULE

The Financials screen provides:

• three-year upload;
• extraction status;
• document detection;
• extracted line items;
• verification confidence;
• discrepancy review;
• source-page reference; and
• final data approval.

This module visually demonstrates the transition from raw financial documents to a verified financial dataset.

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41. AI INTERVIEW MODULE

The AI Interview screen displays:

- current question;
- interview progress;
- question category;
- priority;
- reason for asking;
- triggering financial rule;
- extracted financial context;
- answer options;
- optional explanation;
- AI interpretation; and
- category completion.

This ensures that the user can understand **why the AI is asking a particular question**.

---

42. VALUATION MODULE

The Valuations screen includes:

- indicative valuation range;
- central estimate;
- valuation confidence;
- DCF;
- Market Multiple;
- Adjusted NAV;
- methodology weighting;
- key assumptions;
- sensitivity analysis;
- risks; and
- previous valuation runs.

---

43. ETHICAL AND PROFESSIONAL CONSIDERATIONS

The project deliberately avoids presenting AI as a replacement for professional judgement.

Important principles include:

- transparency;
- explainability;

• traceability;
• data validation;
• user approval;
• reproducibility;
• confidentiality; and
• clear limitation of scope.

The generated report should expressly clarify that CompanyVal AI provides an:

> **AI-assisted indicative valuation simulation based on supplied financial information and assumptions.**

It should not be represented as a statutory valuation or professional certification.

---

44. LIMITATIONS

Like any financial technology prototype, CompanyVal AI has limitations.

44.1 Document Variability

Financial statements may use widely varying layouts and terminology.

44.2 Scanned Documents

Poor-quality scanned documents may reduce extraction confidence.

44.3 Incomplete Financial Information

Valuation quality depends on the information supplied.

44.4 Forecast Uncertainty

Future financial assumptions inherently involve uncertainty.

44.5 Market Benchmark Availability

Market-multiple valuation requires suitable comparable data.

44.6 Qualitative Responses

The system relies partly upon information supplied by the user.

44.7 AI Interpretation

Generative AI can potentially misinterpret information, which is why authoritative calculations and approvals are kept outside the AI layer.

44.8 Indicative Nature

The application does not substitute for specialised professional valuation procedures required for statutory or regulated purposes.

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45. RISK MITIGATION

Risk	CompanyVal AI Control
Incorrect PDF extraction	Python + visual AI verification
AI hallucinated amount	AI cannot overwrite financial data
Incorrect assumption	Human approval required
AI calculation error	Python valuation engine
Unsupported multiple	Explicit configured/user source
Untraceable figure	Source-page audit trail
Sensitive API credential	Backend encryption and masking
AI service failure	Core deterministic engine remains independent
Invalid DCF assumptions	Input validation
Excessive questioning	Priority-based interview

---

46. TESTING STRATEGY

Testing should cover multiple layers.

46.1 Financial Calculation Tests

Validate:

- CAGR;
- EBITDA Margin;
- NOPAT;

- FCFF;
- present values;
- terminal value;
- enterprise value;
- equity value;
- market multiples;
- Adjusted NAV;
- weighted central estimate.

46.2 Extraction Tests

Validate:

- Indian number formatting;
- lakh/crore conversion;
- parentheses;
- negative amounts;
- comparative columns;
- blank cells;
- multiple financial years.

46.3 Rules Testing

Ensure financial conditions trigger the correct interview topics.

46.4 AI Interview Testing

Confirm that:

- questions are contextual;
- prior answers influence follow-up questions;
- questions explain their trigger;
- unnecessary questions are avoided.

46.5 Scenario Testing

Confirm that changes in:

- growth;
- margins;
- WACC;
- terminal growth; and
- multiples

cause mathematically correct changes to valuation.

46.6 Security Testing

Confirm that:

- API keys never appear in frontend source;
- stored keys remain masked;
- keys are not written to logs;
- files are validated before processing.

---

47. DEMONSTRATION CASE

A fictional company such as:

ABC Food Pvt. Ltd.

can be used as the controlled demonstration case.

The demo should contain three years of realistic financial statements and selected valuation characteristics.

Suggested characteristics include:

- meaningful revenue growth;
- improving EBITDA margin;
- a material one-time order;
- increasing receivable days;
- customer concentration;
- manageable financial leverage;
- planned expansion capital expenditure.

This creates enough complexity for the AI Interview and valuation engine to demonstrate genuine value.

48. EXPECTED DEMONSTRATION FLOW

During the capstone demonstration:

Step 1

Create the ABC Food Pvt. Ltd. valuation case.

Step 2

Upload three years of financial statements.

Step 3

Show Python extracting data.

Step 4

Show AI visual verification.

Step 5

Open a deliberate extraction discrepancy.

Step 6

Approve and lock historical numbers.

Step 7

Show financial ratios.

Step 8

Allow the rules engine to detect abnormal revenue growth.

Step 9

AI asks why revenue grew.

Step 10

User specifies that growth partly arose from a one-time order.

Step 11

AI asks an intelligent follow-up.

Step 12

Approve resulting assumptions.

Step 13

Generate DCF, Market Multiple and NAV valuations.

Step 14

Open Simulation Lab.

Step 15

Change WACC or growth assumptions and observe immediate valuation movement.

Step 16

Open **Why this valuation?**

Step 17

Generate the professional report.

This demonstrates the entire AI-assisted workflow in a concise manner.

---

49. AI COMPETENCIES DEMONSTRATED

The project demonstrates several practical AI competencies.

Prompt Engineering

Separate prompts are used for:

- document verification;
- question generation;
- answer interpretation;
- risk analysis; and
- report preparation.

Structured AI Output

AI responses are requested using predefined schemas where appropriate.

Multimodal AI

Rendered financial-statement pages are analysed visually.

AI Orchestration

AI output is integrated into a larger deterministic application workflow.

Human-in-the-Loop AI

The user retains authority over:

- historical figures;
- adjustments;
- assumptions; and
- valuation inputs.

Explainability

The system explains:

- why questions are asked;
- where financial figures came from;
- why a valuation changed; and
- what factors drive value.

---

50. WHY AI IS REQUIRED IN THIS PROJECT

A normal rules engine could calculate a valuation after receiving all necessary inputs.

However, AI materially improves the process where interpretation is required.

For example:

A rules engine may identify:

```
Revenue Growth = 44%
```

but a generative AI system can ask:

> **What primarily caused this growth, and to what extent do you expect that factor to recur?**

It can then interpret:

> "We received a one-time government contract worth approximately ₹2.5 crore which is unlikely to recur."

into structured valuation-relevant information such as:

```
Growth Source        = Non-recurring
Amount               = ₹2.5 Cr
Recurrence           = Low
Normalisation Review = Required
```

This is where generative AI creates value beyond conventional software.

---

51. WHY AI DOES NOT CALCULATE THE FINAL VALUE

Financial valuation should remain:

- reproducible;
- auditable;
- explainable; and
- mathematically deterministic.

Asking an LLM to independently calculate the entire valuation may lead to:

- different results on repeated runs;
- arithmetic mistakes;
- unsupported assumptions; and
- difficulty reconciling outputs.

CompanyVal AI therefore uses the principle:

> **AI understands and explains. Python calculates. Human approves.**

This is the core design philosophy of the project.

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52. FUTURE SCOPE

CompanyVal AI can subsequently be expanded significantly.

Possible enhancements include:

Automatic Comparable Company Data

Integration with authorised financial-market datasets.

Industry Benchmarking

Compare:

- margins;
- growth;
- ROCE;
- leverage; and
- valuation multiples.

Comparable Transaction Analysis

Add precedent transaction methodology.

Startup Valuation

Introduce methods suited to early-stage businesses.

SaaS Valuation

Analyse:

- ARR;
- MRR;
- churn;
- CAC;
- LTV;
- Rule of 40.

Sector-Specific Models

Create specialised valuation engines for:

- manufacturing;
- FMCG;
- SaaS;

- professional services;
- real estate;
- healthcare.

Management Forecast Upload

Analyse budget/projection files.

Bank Statement Verification

Cross-check reported financial activity against banking information.

ERP Integration

Import verified accounting data directly from accounting systems.

Collaboration

Allow management, finance professionals and reviewers to work on a valuation case together.

AI Valuation Assistant

Enable natural-language questions such as:

> Why did the DCF valuation fall after I increased revenue growth?

while answering from the deterministic calculation trail.

---

53. EXPECTED BENEFITS

CompanyVal AI can demonstrate how AI may make preliminary business valuation:

- faster;
- more interactive;
- more explainable;
- more consistent;
- more auditable; and
- easier to understand.

The project also demonstrates that responsible financial AI does not require handing complete control to the model.

Instead, AI can be integrated within professional workflows while preserving deterministic calculations and human judgement.

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54. PROJECT OUTCOME

The intended output is a functional application through which a user can:

> **Create a valuation case → upload three years of financial statements → extract and verify data → complete an intelligent AI interview → approve assumptions → calculate business value using multiple methodologies → perform scenario simulations → understand the key value drivers → generate a professional AI-assisted valuation report.**

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55. CONCLUSION

CompanyVal AI demonstrates a practical application of Artificial Intelligence in the field of accounting and finance.

Rather than treating AI as an autonomous valuation expert, the solution combines the respective strengths of:

- Python;
- deterministic financial rules;
- multimodal AI;
- generative AI;
- valuation methodologies; and
- professional human judgement.

Financial information is first extracted and validated. Rules then identify unusual or important financial characteristics. AI collects additional contextual information through an adaptive interview. Explicit assumptions are subsequently applied to deterministic valuation methodologies. Scenario modelling provides an understanding of valuation sensitivity, while AI converts structured results into understandable insights and a professional narrative.

The fundamental design philosophy of CompanyVal AI can therefore be summarised as:

> ## AI UNDERSTANDS. PYTHON CALCULATES. HUMAN APPROVES.

This separation creates a system that is more transparent, auditable and explainable than an application in which a generative AI model independently determines business value.

CompanyVal AI therefore illustrates how Artificial Intelligence can support professional financial decision-making while retaining appropriate human oversight and mathematical control.

APPENDIX A — CORE MODULES

Module	Principal Function
Dashboard	Portfolio overview
New Valuation	Create valuation case
Financials	Upload, extract and verify
AI Interview	Adaptive qualitative assessment
Valuations	DCF, Multiples and NAV
Simulation Lab	Scenario modelling
AI Insights	Explainable qualitative analysis
Reports	Professional report generation
Settings	AI and application configuration

APPENDIX B — AI RESPONSIBILITY MATRIX

Activity	Python / Rules	AI	Human
PDF technical extraction	✓		
Visual page verification		✓	
Resolve discrepancy		Suggest	✓
Financial ratios	✓		
Trigger financial rules	✓		
Draft contextual question		✓	
Answer business question			✓
Interpret narrative answer	Rules/Schema	✓	Approve where material
Recommend assumption		✓	
Approve assumption			✓
DCF calculation	✓		
Multiple valuation	✓		
NAV calculation	✓		

Activity	Python / Rules	AI	Human
Sensitivity calculation	✓		
Explain valuation	Data	✓	
Generate narrative report	Structured data	✓	Review

APPENDIX C — PROJECT DESIGN PRINCIPLES

1. **Never trust one extraction route where verification is possible.**
2. **Never let generative AI silently alter historical financial information.**
3. **Never let AI invent market assumptions.**
4. **Keep authoritative financial mathematics deterministic.**
5. **Explain why every material AI question is being asked.**
6. **Require explicit user approval for material assumptions.**
7. **Maintain traceability from valuation output to input data.**
8. **Prefer valuation ranges over artificial precision.**
9. **Disclose methodology and assumptions.**
10. **Use AI to augment professional judgement rather than replace it.**

APPENDIX D — PROJECT SUMMARY FOR EVALUATOR

Project: CompanyVal AI — AI-Assisted Business Valuation

Business Problem: Traditional valuation requires extensive financial analysis and qualitative business understanding.

AI Application: Multimodal verification, intelligent Q&A, qualitative interpretation and professional narrative generation.

Python Application: PDF extraction, financial standardisation, accounting validation, ratios, valuation, simulation and sensitivity analysis.

Valuation Approaches: DCF, Market Multiple and Adjusted NAV.

Human Oversight: Required for financial corrections, normalisation and assumptions.

Key Innovation: Dual financial-document verification combined with an adaptive AI interview and deterministic valuation engine.

Core Philosophy:

AI Understands. Python Calculates. Human Approves.